

Autism Spectrum Disorder Classification System

A Comparative Study of Supervised and Unsupervised Machine Learning Models

Machine Learning Project Report

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Project Repository: <https://github.com/AhmedNadeem-ai/Autism-Spectrum-Disorder-Classification-System>

Live Streamlit App: <https://autism-spectrum-disorder-classification-system.streamlit.app/>

Introduction

Autism spectrum disorder (ASD) is a lifelong neurodevelopmental condition that affects communication, behavior, and social interaction. It presents differently from person to person, making its identification and management complex. Because autism is common and often diagnosed later than ideal, early detections supported by clinical practice and modern machine learning methods have significant value for individuals, families, and society.

Understanding The Disorder

Autism spectrum disorder refers to a broad range of differences characterized by challenges in social communication and the presence of repetitive or restricted behaviors. Autism Speaks explains that ASD varies widely, with some individuals speaking fluently while others are nonverbal, and with cognitive abilities ranging from intellectual disability to high intelligence (Autism Speaks, n.d.).

Symptoms

Common features include difficulty with social interaction, differences in communication, repetitive movements or routines, and strong preferences for specific activities or sensory experiences. Signs usually appear by age two or three. The DSM-5-TR organizes ASD into three levels based on support needs: level 1 (requiring support), level 2 (substantial support), and level 3 (very substantial support). Co-occurring conditions such as ADHD, anxiety, depression, gastrointestinal problems, seizures, and sleep disturbances are also common (Autism Speaks, n.d.; Mayo Clinic Staff, 2024).

Prevalence And Impact

Autism affects approximately 1 in 31 children and 1 in 45 adults in the United States, according to Autism Speaks (n.d.). Boys are diagnosed about four times more often than girls, partly because autistic traits in girls may appear more subtle, leading to underdiagnosis or

misdiagnosis. Autism can affect educational progress, social development, and long-term independence, though outcomes improve significantly when individuals receive timely support.

Why Detecting This Disorder is Important

Early detection of autism spectrum disorder is important because it supports children academically, clinically, and socially. Identifying ASD early allows access to specialized educational support, individualized learning plans, and therapies that strengthen communication and reduce long-term academic barriers. Clinically, early diagnosis enables healthcare providers to introduce interventions during critical developmental periods, improving communication, daily functioning, behavioral outcomes, and long-term independence (Mayo Clinic Staff, 2024). On a societal level, timely detection reduces long-term care costs, supports families and community systems, promotes inclusion, and helps autistic individuals participate more fully in education, employment, and social life.

Importance of Machine Learning for Disorder Detection

Machine learning supports early ASD detection by analyzing behavioral patterns, speech features, facial expressions, screening questionnaires, and large datasets more efficiently than manual observation alone. ML models can identify early developmental differences, assist clinicians in diagnosis, reduce misclassification, and help close gaps in early identification, especially in settings where specialists are limited. These technologies do not replace clinicians but act as support tools to improve accuracy and access.

Assigned Datasets Information

For this project, the dataset used is **ASD Children Traits dataset** from Kaggle (Uppulurimadhuri, n.d.). This dataset contains behavioral and demographic data related to autism traits in children.

Dataset Information

The primary dataset used for this project is the **ASD Children Traits Dataset** sourced from Kaggle (Uppulurimadhuri, n.d.), originally collected by researchers at the **University of Arkansas (Computer Science Department)**. The dataset was compiled to analyze early developmental indicators and predict whether a child is likely to develop Autism Spectrum Disorder (ASD) traits.

Dataset Specifications

- **Dataset Dimensions:** 1,985 rows (records) and 28 columns (features).
- **Data Types:** Mixed data types including binary indicators (0/1), quantitative scores/ages, and categorical demographic variables.
- **Target Variable:** The final column serves as the binary target class (0 or 1), indicating whether the individual is projected to develop ASD traits

Key Features & Attributes

The 28 attributes encompass screening questionnaires, clinical indicators, and demographic background details:

- **Screening Scores & Standardized Instruments:**
 - **Autism Spectrum Quotient (AQ-10 / Columns A1–A10):** A 10-item self-scoring questionnaire where scores range from 0 to 10. A score of 6 or higher serves as the clinical threshold indicating significant autistic traits.
 - **Qchat_10_Score:** Quantitative score from the Quantitative Checklist for Autism in Toddlers.
 - **Childhood Autism Rating Scale (CARS):** Standardized behavioral observation score.
 - **Social Responsiveness Scale (SRS):** Quantitative measure of social impairment and communication traits.

- **Clinical & Developmental Co-occurrences:**
 - Speech Delay / Language Disorder
 - Global Developmental Delay / Intellectual Disability
 - Learning Disorders & Social/Behavioral Issues
 - Genetic Disorders
 - Mental Health Factors (Depression, Anxiety Disorder)
- **Demographics & Medical History:**
 - **Individual Demographics:** Age (in years), Sex, Ethnicity.
 - **Early Indicators:** History of neonatal jaundice, Family member with ASD.

Methodology and Data Preprocessing

Before training the machine learning models, the raw dataset underwent a rigorous preprocessing pipeline to ensure optimal model performance and data integrity:

- **Missing Value Imputation:** Missing entries were addressed using feature-wise mean imputation to preserve dataset size without skewing feature distributions.
- **Feature Selection:** Non-essential demographic features (such as Ethnicity and Neonatal Jaundice) were excluded to reduce noise and optimize the models for clinical and behavioral indicators.
- **Categorical Encoding:** Non-numeric categorical attributes were converted into a machine-readable format using one-hot/dummy encoding.
- **Feature Scaling:** Numerical features were standardized using StandardScaler, bringing all attributes to a uniform scale, which is particularly critical for distance-based models like KNN and SVM.

- **Dimensionality Reduction:** Principal Component Analysis (PCA) was applied to compress the feature space for the unsupervised clustering models (reducing to 2 components), allowing for optimal cluster separation and 2D visualization.

System Implementation: Interactive Web Interface

To bridge the gap between theoretical machine learning and practical clinical application, an interactive web interface was developed and deployed using the **Streamlit** framework in Python.

Rather than limiting the project to static model evaluations, the deployed system allows users to interact dynamically with the trained models. The application processes real-time user inputs through the exact same scaling and transformation pipelines used during training.

Key Features of the GUI:

- **Comprehensive Data Intake:** The interface collects demographic details, clinical assessment scores (e.g., QCHAT-10, CARS, SRS), co-occurring conditions, and automatically scores a full 10-question Autism Spectrum Quotient (AQ-10) behavioral assessment.
- **Multi-Model Diagnostic Evaluation:** Upon execution, the system feeds the user's data into all saved (pickled) supervised and unsupervised models simultaneously.
- **Consensus and Confidence Tracking:** The application outputs the prediction of the top-performing model (Decision Tree) alongside its confidence probability. Furthermore, it calculates a "Consensus Agreement," showing how many of the deployed models independently arrived at the same diagnostic conclusion.

Screenshots:

ASD Traits Prediction System

Ask Gemini

School

Deploy

Autism Spectrum Disorder (ASD) Traits Predictor

This clinical decision support tool utilizes trained Machine Learning models to evaluate behavioral, screening, and demographic indicators for early ASD trait screening.

System Status: 10 / 10 models loaded

Patient Information & Clinical Indicators

Fields marked with an asterisk (*) are required for analysis.

Demographics & History

Age (Years, 1-18) *

Sex *

Male Female

Ethnicity

Choose an option

Clinical Assessments

QCHAT-10 Score (0-10) *

Childhood Autism Rating Scale (1-4) *

Social Responsiveness Scale (0-10) *

Co-occurring Conditions

Genetic Disorders? *

Yes No

Global Developmental Delay? *

Yes No

Social / Behavioral Issues? *

Yes No

ASD Traits Prediction System

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Autism Spectrum Quotient (AQ-10) *

A1: Does not easily respond to social cues or unspoken meanings

Yes No

A6: Finds it difficult to understand other people's feelings

Yes No

A2: Strongly prefers repetitive routines and is unsettled by changes

Yes No

A7: Experiences strong sensitivities to everyday sensory inputs (sounds, lights, textures)

Yes No

A3: Struggles to understand the social intentions behind what people say

Yes No

A8: Shows intense, restricted focus on specific topics of interest

Yes No

A4: Focuses heavily on small details, patterns, or visual elements that others miss

Yes No

A9: Struggles with the natural back-and-forth flow of conversation

Yes No

A5: Has an unusual speech rhythm, tone, or intonation

Yes No

A10: Finds unexpected changes to plans or schedules very difficult to handle

Yes No

Run Multi Model Diagnostic Evaluation

ASD Traits Prediction System

Ask Gemini

School

Deploy

Evaluation Results

PRIMARY ASSESSMENT

No ASD Traits Detected

Decision Tree Model

CONFIDENCE INTERVAL

52.6%

Probability Score

CONSENSUS AGREEMENT

ASD Traits Detected

6 out of 10 models aligned

Comprehensive Model Analysis

Model Architecture	Diagnostic Prediction	Confidence Score	Baseline Accuracy
Decision Tree (Top Model)	No ASD Traits Detected	52.6%	95.90%
K-Nearest Neighbors (KNN)	ASD Traits Detected	100.0%	93.69%
Support Vector Machine (SVM)	ASD Traits Detected	N/A	93.18%
Logistic Regression (L1)	ASD Traits Detected	100.0%	91.16%
Logistic Regression (L2)	ASD Traits Detected	100.0%	88.38%
Logistic Regression (Base)	ASD Traits Detected	100.0%	88.38%
Gaussian Naive Bayes	ASD Traits Detected	100.0%	71.45%
K-Means (k=2)	No ASD Traits Detected	N/A	66.15%
Gaussian Mixture Model	No ASD Traits Detected	100.0%	66.15%

Results and Evaluation

Supervised Learning Models

Seven supervised machine learning models were implemented and evaluated on the ASD Children Traits dataset. Performance was measured using Accuracy and F1-Score to compare classification efficacy.

Model Name	Accuracy	F1-Score
Decision Tree	95.96%	0.9623
K-Nearest Neighbors (KNN)	93.69%	0.9406
Support Vector Machine (SVM)	93.18%	0.9343
L1 Logistic Regression	91.16%	0.9176
Logistic Regression	88.38%	0.8905
L2 Logistic Regression	88.38%	0.8905
Gaussian Naive Bayes	71.46%	0.7139

Analysis of Supervised Performance:

The **Decision Tree** model emerged as the top-performing classifier, achieving the highest overall Accuracy of **95.96%** and an F1-Score of **0.9623**. Instance-based and margin-based models like KNN (93.69%) and SVM (93.18%) also demonstrated strong predictive capability. Feature selection via L1 regularization improved Logistic Regression performance from 88.38% to 91.16%. Gaussian Naive Bayes achieved the lowest metrics (71.46%), likely due to its strict assumption of feature independence.

Unsupervised Learning Models

Five unsupervised clustering algorithms were evaluated with $k = 2$ clusters to assess how naturally the dataset groups into binary risk categories without ground-truth labels during training.

Model Name	Accuracy	F1-Score
K-Means ($k = 2$)	66.15%	0.6842
Agglomerative ($k = 2$)	66.15%	0.6842
Divisive Approx ($k = 2$)	66.15%	0.6842
Gaussian Mixture ($k = 2$)	66.15%	0.6842
Fuzzy C-Means (Test Set)	65.24%	0.6584

Analysis of Unsupervised Performance:

The centroid-based, hierarchical, and probabilistic clustering methods (K-Means, Agglomerative, Divisive, and Gaussian Mixture) all achieved identical top performances of **66.15% Accuracy** and an **F1-Score of 0.6842**. Fuzzy C-Means trailed slightly at 65.24%.

Key Takeaway

Supervised learning models significantly outperformed unsupervised clustering approaches on this dataset (95.96% vs. 66.15% max accuracy). This confirms that labeled behavioral traits are essential for high-accuracy screening tools, making the **Decision Tree** the clear optimal choice for practical deployment.

References

Autism Speaks. (n.d.). What is autism? <https://www.autismspeaks.org/what-autism>

Mayo Clinic Staff. (2024). Autism spectrum disorder. Mayo Clinic. <https://www.mayoclinic.org>

Uppulurimadhuri. (n.d.). *ASD children traits* [Dataset]. Kaggle.

<https://www.kaggle.com/datasets/uppulurimadhuri/dataset>